

## CORE PLATFORM

## OPTIONAL ADD-ON

## KEY TECHNOLOGIES

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### PRESENTATION LAYER

(User Interface)

#### React 18 + TypeScript Web Application

Pages • Components • State Management

Wagmi / Viem • Wallet Integration

#### AI Agent Interface

(Qwen3-8B)

#### MCP Tool Server

(17 Banking Tools)

React 18, TypeScript,  
Wagmi, Viem,  
Ethers.js

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### SMART-CONTRACT LAYER

(On-Chain – EVM)

#### 15 Modular Smart Contracts (Solidity)

##### Core Four-Tier Hierarchy

World Bank Reserve (Tier 1)



National Banks (Tier 2)



Local Banks (Tier 3)

##### 12 Planned Modules

- Loans, SavingsVault, FixedDeposit
- GroupLendingPool, InterBankPool
- CreditPassport, Compliance (ZKP-KYC)
- Fraud Detection, Oracles, Automation
- and others

Solidity ^0.8.x  
OpenZeppelin,  
EVM (zkEVM)

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### OFF-CHAIN SERVICES LAYER

(Backend & Data Services)

#### Express.js REST API

(Node.js)

#### PostgreSQL Relational Database

#### FastAPI ML Scoring Service

#### Chainlink Oracle Integration

#### Event Listener (Indexer)

Node.js, Express.js,  
PostgreSQL, FastAPI,  
Redis, WebSockets

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### CHAINLINK INFRASTRUCTURE

(Oracle & Automation)

#### Chainlink Decentralized Oracle Services

##### Chainlink Functions

(Off-Chain Compute)

##### Chainlink Automation

(Scheduled Tasks)

##### Chainlink Price Feeds

(Market Data)

##### Chainlink Proof of Reserve

(Reserve Verification)

Chainlink Functions,  
Automation, Price  
Feeds, Proof of  
Reserve